

Set 24: Interpretation of formulae

The formula of a substance indicates the relative number of atoms of each element in the substance. For example:

One molecule of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, contains 6 atoms of carbon, 12 atoms of hydrogen and 6 atoms of oxygen.

Also one mole of glucose molecules, $\text{C}_6\text{H}_{12}\text{O}_6$ contains 6 mol of carbon atoms, 12 mol of hydrogen atoms and 6 mol of oxygen atoms.

Notes

Examples

1. How many moles of oxygen atoms are contained in 5.00 moles of calcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$) formula units?

$$\begin{aligned}n(\text{oxygen atoms}) &= 8 \times n(\text{Ca}_3(\text{PO}_4)_2) \\&= (8) (5.00)\end{aligned}$$

$$n(\text{oxygen atoms}) = 40.0 \text{ mol}$$

2. What mass of oxygen atoms is contained in 132 g of carbon dioxide (CO_2)?

(a) Find the number of moles of CO_2 using $n = \frac{m}{M}$ $M(\text{CO}_2) = 44.01 \text{ g mol}^{-1}$

$$n(\text{CO}_2) = \frac{m}{M}$$

$$= \frac{132}{44.01}$$

$$n(\text{CO}_2) = 3.00 \text{ mol}$$

- (b) Find the number of moles of oxygen from the number of moles of CO_2

$$n(\text{O}) = 2 \times n(\text{CO}_2)$$

$$= (2) (3.00)$$

$$n(\text{O}) = 6.00 \text{ mol}$$

- (c) Find the mass of O using $n = \frac{m}{M}$ $M(\text{O}) = (16.0 \text{ g mol}^{-1})$

$$n(\text{O}) = \frac{m}{M}$$

$$m(\text{O}) = nM$$

$$= (6.00) (16.0)$$

$$m(\text{O}) = 96.0 \text{ g}$$

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Notes

Set 24: Exercises

- Calculate the number of moles of:
 - hydrogen atoms in 2.50 moles of calcium hydroxide ($\text{Ca}(\text{OH})_2$) formula units
 - nitrogen atoms in 0.0500 moles of ammonium nitrate (NH_4NO_3) formula units
- Calculate the number of moles of:
 - phosphorus trichloride (PCl_3) molecules which contain 21.0 moles of chlorine atoms
 - potassium permanganate (KMnO_4) formula units which contain 2.00 moles of oxygen atoms
- Calculate the mass of:
 - oxygen in 795 g of copper(II) oxide (CuO)
 - potassium in 1.04 g of potassium sulfate (K_2SO_4)
 - calcium in 38.4 g of calcium oxalate (CaC_2O_4)
- What mass of:
 - sulfur dioxide (SO_2) contains 193 g of sulfur?
 - ammonium sulfate ($(\text{NH}_4)_2\text{SO}_4$) contains 0.0960 g of hydrogen?
 - octane (C_8H_{18}) contains 36.0 g of carbon?
- Given 2.50 moles of ammonium phosphate ($(\text{NH}_4)_3\text{PO}_4$) formula units, calculate the number of moles of:
 - ammonium (NH_4^+) ions
 - phosphate (PO_4^{3-}) ions
 - nitrogen atoms
 - hydrogen atoms
 - phosphorus atoms
 - oxygen atoms
- Calculate the mass of iron in 6.40×10^2 g of iron(III) oxide.
- What mass of ethanol ($\text{C}_2\text{H}_5\text{OH}$) contains 144 g of carbon?
- What mass of oxygen is contained in 2.50 moles of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$)?
- How many moles of nitrogen atoms are contained in 1.80×10^4 g of urea ($\text{CO}(\text{NH}_2)_2$)?
- Calculate the mass of soap, sodium stearate ($\text{NaC}_{17}\text{H}_{35}\text{COO}$), that contains 1.25 g of carbon.